State-Visitation Failures in On-Policy Distillation: An Audited Agent Case Study and Confirmatory Protocol

Anonymous ACL submission

Abstract

On-policy distillation supervises states visited by the student, yet a long-horizon agent may rarely reach persistent states where its teacher has a measurable conditional advantage. We audit this failure mode for a Qwen3.5-2B agent acting through 17 typed tools in an open-source MMORPG. In one reported six-hour run per arm, natural-visitation distillation changes tool use and interaction tempo but leaves quest score at 12/30. A later round adds student rollouts from a hand-selected, loadable intermediate player state whose legal reachability was not certified. Evaluated from a reported canonical start, without intermediate-state initialization, the resulting policy scores 15/30 and all three clustered prompt variants pass a wall that none passes in the earlier runs. This sequence is descriptive, not a causal estimate: training rounds differ in more than state source, and exact historical execution provenance is incomplete. We therefore specify a falsifiable study that selects training-only states using frozen visitation, teacher-advantage, validity, reachability, and recoverability criteria; crosses player state with model-visible history; and compares natural OPD, generic resets, successful prefixes, and turn-level guidance under matched budgets. A separate nine-cell local pilot validates the prospective execution path, but none of its five-minute cells advances a quest. The contribution is an audited case study and bounded protocol, not an established method effect.

1 Introduction

Post-training a small language model to act over long horizons is not just a token-prediction problem. The model chooses which environment states become training data. If it never reaches a state where its teacher knows what to do, dense teacher feedback is unavailable exactly where it is needed. This creates a state-visitation bottleneck: improving token-level agreement on common states can alter style without transferring the competence that distinguishes the teacher.

We examine that bottleneck in a persistent tool-use setting. A Qwen/Qwen3.5-2B student (Qwen Team, 2026) plays a tile-based MMORPG through a typed action interface. The world persists across conversation rollovers, and benchmark progress requires long chains of gathering, navigation, dialogue, and quest actions. A scaffolded 4B Qwen/Qwen3.5-4B teacher reaches states that the 2B student does not. We apply reverse-KL on-policy distillation to the student's own emitted tokens. The historical comparison contrasts natural student visitation with a mixture that begins some rollouts from loadable, walkability-checked intermediate database states; the confirmatory protocol additionally freezes a measurable state-selection rule before training.

The present evidence is hypothesis-generating and under-replicated. It contains one complete six-hour run per principal arm; the three agents within a run are prompt variants sharing weights, harness, and launch conditions, not independent experimental units. We therefore separate observations from claims throughout. The observed sequence is: base 12/30, natural-visitation round 12/30, and state-augmented round 15/30. In the last arm, every prompt variant passes the same wall from a canonical-start evaluation world with no intermediate-state initialization. This supports the hypothesis that training visitation matters, but it does not isolate state seeding from all round-to-round changes or estimate population-level uncertainty.

This working draft makes three bounded contributions:

1. We audit a state-visitation failure case and separate observations that survive artifact review from causal claims that do not. We provide a confirmatory protocol with controls capable of falsifying the proposed player-state intervention.

2. We distinguish behavioral imitation from competence in the surviving historical record: the first round closely changes measured tool mix and tempo toward the teacher while the benchmark score remains fixed.

3. We document an interface diagnostic that token-average analyses obscure: in one targeted teacher-forcing probe, a malformed continuation reportedly has positive teacher-over-student distillation advantage. This secondary diagnostic does not yet establish a within-context preference reversal and is separated from a runtime recovery result and from any general mechanism claim.

The broader lesson is methodological. Model weights, visited states, and runtime affordances are different interventions. A credible agent-training result must preserve and cross them rather than roll them into one headline.

2 Setting

2.1 Persistent tool-use benchmark

The environment is a pinned fork of Kaetram, an open-source tile-based MMORPG (Kaetram contributors, 2026), backed by a game server and MongoDB. Each policy controls three agents with different priority prompts (grinder, completionist, and explorer) through 17 typed tools. Tools cover observation, navigation, combat, gathering, dialogue, quest inspection, inventory use, and crafting. The prompt includes procedural quest guidance, so the benchmark measures execution of a supplied plan rather than autonomous world-model discovery.

The primary development metric, CORE-3, counts newly reached stages in three quest chains: Foresting (3 stages), Herbalist's Desperation (3), and Rick's Roll (4). Each agent can gain at most ten stages; a three-agent run therefore scores in $[0, 30]$. Historical notes describe runs as beginning from a freshly initialized player state in a restarted server, but no fail-closed reset receipt or complete shared-world hash survives. Runs continue for six hours while conversation sessions roll over under a fixed context budget and game state persists. The intended confirmatory experimental unit is a full run with an attested reset. Prompt variants are clustered observations within that run.

2.2 Agent and interface contract

The policy acts in an observe–reason–act loop through an OpenAI-compatible model endpoint and a typed tool server. Navigation and selected interactions contain deterministic affordances such as path finding and automatic walking; the model still decides goals, tools, and arguments. The current development record contains an important interface asymmetry: older base-model runs used a native model-visible tool schema while some fine-tuned runs did not receive an identical serialization. We therefore do not use those comparisons as primary evidence. The confirmatory implementation requires normalized full-schema hash parity across training and every evaluation arm.

3 On-Policy Distillation Under Visitation Constraints

Let $x \in \mathcal{X}$ denote persistent player state (including position, inventory, skills, and quest progress) and $h \in \mathcal{H}$ the model-visible interaction history. The joint occupancy state is their joint value $z = (x, h)$; two rollouts with the same database snapshot but different visible histories need not induce the same action distribution. Let μ initialize player states, $q(h \mid x)$ initialize visible histories, and $d^{\pi}_{\mu,q}(x, h)$ denote the joint occupancy induced thereafter by policy $\pi(a \mid h)$ and the environment dynamics, whose transitions depend on the latent player and world state. The model does not observe x directly; it acts only from h, which may contain tool observations derived from x. The canonical serving initializers are μ_0 and q_0. A reverse-KL on-policy objective can be written

$$\mathcal{L}_{\mathrm{OPD}} = \mathbb{E}_{(x,h)\sim d^{\pi_S}_{\mu,q}} \mathbb{E}_{a\sim\pi_S(\cdot|h)} \left[\log \pi_S(a \mid h) - \log \pi_T(a \mid h)\right]. \quad (1)$$

The teacher scores the student's emitted tokens. Training uses a clipped importance-weighted update around the rollout policy, with advantages frozen at data-build time. Full implementation details appear in Appendix A.

Equation 1 makes the bottleneck explicit. If a teacher advantage is concentrated on a set $H \subseteq \mathcal{X} \times \mathcal{H}$ for which $\Pr_{(x,h)\sim d^{\pi_S}_{\mu_0,q_0}}[(x,h) \in H] \approx 0$, token-dense grading supplies almost no information about that advantage. We change the state source, not the loss: some student rollouts begin from intermediate player states, and their records are mixed with natural student rollouts. The student still acts and the same teacher still grades every emitted token. Because changing x can also change the compatible h, the confirmatory design crosses player-state initialization with matched history construction rather than treating a database snapshot as the complete policy state.

The seeded states are typed persistent player states rather than textual demonstrations. Player rows specify quest stage, skill level, inventory, and a walkability-checked position. End-to-end loadability is necessary but not sufficient for legal reachability. Each confirmatory candidate must also carry either a hash-verified witness trajectory from the canonical start that is replayed by a pinned checker, or an invariant certificate executed by a pinned checker that establishes a valid path to the snapshot. Hash continuity or certificate-shaped metadata alone is not sufficient. Evaluation always starts from the canonical initial player state; prospective runs nevertheless record distinct gameplay-RNG seeds. This resembles reset-based curricula in reinforcement learning (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). TCOD uses teacher-success prefixes to initialize multi-turn OPD curricula (Wang et al., 2026); Guided-OPD mixes teacher and student turns and anneals guidance to zero (Li et al., 2026a); ReOPD treats prefix selection as a teacher-reliability problem (Liao et al., 2026). These works preempt generic novelty for intermediate starts or state-distribution design. Our narrower object is a direct, prefix-independent player-state initializer. A candidate state need not lie on a successful teacher trajectory and is selected by a frozen rule rather than by progress alone.

Table 1: Frozen targeted-state admission rule. L and U are 95% Wilson bounds recomputed from hash-bound Boolean trials. Every row must pass.

Criterion	$n_{\min}$	Rule
Visit rate	500	$U(v) \leq .05$
Teacher success	100	$L(p_T) \geq .60$
Teacher–student gap	100 each	$L(p_T) - U(p_S) \geq .30$
Recovery	100	$L(p_R) \geq .80$
Validity/reachability	–	pinned check

Let $v(x)$ be estimated natural student visitation, and let $p_T(x,q)$ and $p_S(x,q)$ be repeated teacher and student success rates under a frozen history constructor q. The confirmatory selector admits a player state only if it passes player-state and compatibility invariants, is unfinished and task-relevant, is recoverable, satisfies $v(x) \leq \tau_v$, has certified reachability, and satisfies $p_T(x,q) - p_S(x,q) \geq \tau_\Delta$ together with a minimum teacher success threshold. Counts and denominators, source-run identifiers, complete snapshots, and hash-verified validation artifacts are frozen before outcomes. This rule distinguishes the proposed method from the hand-selected milestone used in the exploratory round.

4 Evidence Protocol

We grade evidence before interpreting it. For a fixed checkpoint, a fresh-world evaluation run is the independent unit; agents within it share the policy, launch, infrastructure, and analysis pipeline. For a training-method claim, the independent unit is instead a freshly initialized training seed/checkpoint, with evaluation seeds nested within it. Exact scores from single historical runs are descriptive. Counts over agents or sessions diagnose mechanisms but cannot be treated as independent replications of either effect.

The principal historical sequence uses a 2B model lineage, six-hour canonical-start evaluation protocol, benchmark, and nominal harness, but it is not a same-checkpoint comparison. Round 1 trains on natural base-policy rollouts. Round 2 initializes from round 1 and mixes natural round-1 rollouts with 2,326 records collected by the round-1 policy from a loadable, walkability-checked Herbalist milestone state. The round-2 corpus contains 7,849 states, split into 7,024 training and 825 held-out records.

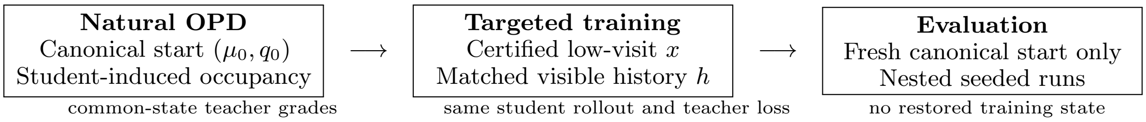

Figure 1: Intervention boundary. Targeted player-state restoration changes training occupancy, while the loss and canonical-start evaluation contract stay fixed. Model-visible history is crossed explicitly because a database snapshot alone is not the policy state.

Because round 2 also changes the initialization and data batch, round 1 versus round 2 is not a clean state-augmented-versus-natural-only training ablation.

Separately, before observing outcomes we registered a local feasibility pilot using no paid endpoint and crossing Base, round-2, and round-3 public weight snapshots with three paired inference/environment seeds. Each cell uses one completionist prompt identity, recovery off, a canonical tool schema, the same locked tokenizer and patched render contract, and one five-minute canonical-start episode. This pilot tests execution and action throughput only; it is not the six-hour weights-by-recovery factorial and cannot establish policy superiority.

The *wall* column measures passage of Herbalist stage 1 by the three prompt variants. It is a useful within-run consistency check, not a binomial sample of independent policies. We preserve raw pre-rewrite emissions for confirmatory format analyses; rewritten history is not an authoritative defect counter.

5 Exploratory Results

5.1 Behavior changes without score change in the reported natural-visitation round

Round 1 leaves CORE-3 unchanged at 12/30 and every agent reaches the same 4/10 stage profile as base. Nevertheless, several behavioral measurements move toward the 4B teacher (Table 3). Rounded to one decimal place, the average number of turns per session equals the teacher value; observation and navigation shares approach the teacher, and query frequency overshoots it. The result is consistent with mode-seeking behavioral imitation and with the hypothesis that agreement on frequently visited states can be insufficient for this quest wall.

Development logs also contain per-verb reverse-KL diagnostics, but the surviving summary lacks matched-state opportunities and uncertainty. We therefore make no quantitative KL–behavior correspondence claim from those values. The aggregate tool mix is descriptive and does not establish that a specific token update caused downstream quest behavior.

5.2 The seeded-training round is followed by canonical-start wall passage

The 4B teacher demonstrates competence beyond the Herbalist stage-1 wall. In the reported base and round-1 runs, zero of three prompt variants crosses that wall; the available record does not provide a defensible denominator for a broader natural-visitation rate. Seeded collection starts the round-1 policy at the accepted quest with the required gathering skill and near the target resource. All three seeded collectors pass the wall, creating student-authored actions and teacher scores in this region.

After training on the natural-plus-seeded mixture, round 2 is reported as evaluated from a canonical-start player state in a restarted server, with no intermediate-state initialization and with recovery disabled. Neither a reset receipt nor its historical gameplay-RNG seed was preserved. It scores 15/30, with each prompt variant at 5/10. All three pass Herbalist stage 1, compared with zero of three under both the base and round-1 policies. Logs show earlier quest acceptance, explicit tracking of the skill gate, resource rotation after empty harvests, and less premature return to the quest giver. These are concrete trajectory differences consistent with the intended mechanism.

The valid statement is deliberately narrow: *a seeded-training round was followed by unanimous canonical-start wall passage in one run.* The present data do not show that seeding alone caused the improvement. The required causal experiment trains fresh students from the same checkpoint under identical budgets,

Table 2: Observed development runs and the claims they permit. Scores are descriptive because there is one run per arm. Recovery is a runtime interface affordance.

Policy	Training states	Recovery	CORE-3	Wall	Interpretation
Base 2B	–	off	12	0/3	Reference run
OPD round 1	natural	off	12	0/3	Behavior changed; score did not
OPD round 2	natural + seeded	off	15	3/3	Reported recovery-off observation
Round-2 weights	–	on	17	–	Recovery ablation; one run
Round-3 weights	broader seeded	on	18	–	Weights plus interface; not pure weights

Table 3: Behavior changes in the first OPD round. Percentages are shares of tool calls in one run per policy; they are diagnostics, not independent trials.

Measure	Base	Round 1	Teacher
Turns per session	9.1	6.1	6.1
Observe share	26.6%	35.5%	36.1%
Navigate share	14.3%	9.4%	9.1%
Attack share	17.9%	6.6%	9.1%
Query-quest share	5.3%	18.0%	8.7%
Chars./turn	411	1,248	864

Table 4: Preregistered local feasibility pilot. Each snapshot has three paired, five-minute cells. Calls/min uses observed wall time, including the final in-flight generation. No-call counts include thinking-only generations. These are descriptive diagnostics, not policy-effect estimates.

Weights	Calls/cell	Calls/min	No-call sum
Base	5, 6, 5	0.968	21
Round 2	1, 5, 6	0.736	22
Round 3	5, 4, 5	0.872	21

seeds, and data construction, varying only whether the legally certified state mixture is included (Section 7).

5.3 Weights and recovery solve different failures

Round 3 combines broader state seeding with a runtime recovery affordance and scores 18/30. That number is not a pure-weights result. A separate unmatched run of round-2 weights with recovery reports 17/30. The run directories were recovered externally, but missing configuration parity, checkpoint/render attestation, reset receipts, and gameplay seeds prevent attribution of either score difference to the interface or weights. The reported recovery-off contrast is base 12/30 versus round 2 15/30, one run per arm; exact checkpoint, render, seed, and artifact parity is unavailable, so even this contrast is not a clean causal weights comparison.

The recovery mechanism detects a dropped malformed call, reconstructs the executable call, rewrites the retained history into canonical form, executes the action, and returns an explicit correction note. Historical notes report 405 rewritten sessions with exactly one recovery marker and no later marker. Recovered rewritten bundles exist, but raw pre-rewrite emissions are absent and the 405 count has not been independently regenerated. It therefore does not show that weights learned the format, that the model self-corrected, or that recovery generalizes to other defects.

5.4 Prospective pilot executes all cells, not the task

All nine preregistered local cells completed with exact canonical-start projections and seed-bound environment-RNG receipts. Across 106 raw endpoint generations, 42 emitted one structured call and 64 emitted none (Table 4). Every emitted call had a canonical tool name and JSON-object arguments; the retained stderr logs contain no logged `API error:` line, and every final session records duration exhaustion. Rehashing the 261 files in the sealed per-cell inventories found no mismatch. Mean structured calls per observed minute were 0.968 for Base, 0.736 for round 2, and 0.872 for round 3. The per-cell counts overlap substantially, and with only three short cells per snapshot we perform no superiority test. No cell advanced a quest or a CORE-3 stage. Thus the pilot provides a narrow operational observation: the matched local stack executed the registered paired cells and retained internally hash-checked evidence; five minutes is too short to support an informative task-level estimate.

6 A Teacher-Forcing Failure at the Interface

Round 1 introduces a malformed parameter-key pattern that is absent in the base run. The emitted form places a Boolean value inside the parameter key, so the parser silently drops the intended argument. The training records contain the correct syntax, and historical notes report that the teacher favors the correct delimiter in clean base-policy contexts. After the malformed prefix is already present, the same notes report teacher log probability -0.161 (about 85% probability) for continuing the malformed form, versus -0.253 under the student. This is a positive teacher-over-student distillation advantage for one candidate. The reported correct-form observation uses a different state; without both canonical and malformed candidates scored in each matched context, the record does not establish a preference reversal.

Thus teacher generation and teacher-forced grading are distinct objects, but the historical probe establishes only a candidate failure mode. A teacher that generates canonical syntax from a clean history may still assign a locally useful distillation signal to continuation of a student's defect after that defect enters the context. We call this a *teacher-forcing copy-prior hypothesis*. The current record lacks the paired artifact needed to show that the teacher actively prefers the malformed candidate.

This hypothesis relates to exposure bias (Bengio et al., 2015) and to failures of token-level teacher agreement on degraded prefixes. CCOPD already conditions a teacher on clean, canonical evidence to reduce self-anchored drift in a multi-turn student (Lin et al., 2026); that broad clean-context intervention is not novel here. The narrower tool-syntax hypothesis gives a testable prediction: paired histories differing only in canonical versus malformed prior syntax should change teacher log-odds while holding state and candidate continuations fixed. Section 7 turns that prediction into a multi-state, multi-defect experiment.

7 Confirmatory Study Design

The repository contains a draft fail-closed launcher contract that enumerates the following matched arms. A hash-pinned preparation adapter validates and normalizes every arm's immutable record contract. Stacked drafts additionally freeze one fresh LoRA parameterization across the matrix, provide a direct-token corrected-interface SFT trainer, and prepare the first stage of the SCoRe-style condition. They do not supply verified arm artifacts, a completed SCoRe-style second stage, a live Guided-OPD collector and actor-conditioned objective backend, or accelerator execution. One stacked draft validates the prospective Guided-OPD role schedule and complete mixed-history record contract, then blocks the legacy trainer. Results are not reported until the remaining implementation boundaries, the full protocol, and immutable run bundles exist.

Primary endpoint and unit. The method estimand is over independently initialized training seeds/checkpoints. Within each training seed, evaluation seeds are nested: each evaluation is a six-hour run from a fresh canonical-start world, with no intermediate player-state initialization and a registered gameplay-RNG seed. The preregistered endpoint aggregates total CORE-3 stage gain across the three clustered agents and then across the frozen evaluation-seed schedule. Quest-wall passage is secondary. The current five-training-seed launcher grid is an unpowered execution template, not a justified sample size; no method claim is permitted until a prospective two-level power analysis freezes both training and nested evaluation replication. All runs, confidence intervals, effect sizes, exclusions, and stopping rules must be reported.

State-source ablation. The six implemented primary training contracts are natural OPD, targeted persistent player states, random-valid player states, progress-matched player states, TCOD-B2F teacher-success prefixes, and Guided-OPD. Four prespecified mechanism/baseline arms—visitation-only selection, teacher-advantage-only selection, corrected-interface SFT, and SCoRe-style first-error prefixes—are analyzed separately rather than silently counted as extra primary arms. Match teacher, optimizer, action-token and teacher-scoring budgets, environment interactions, and training-seed schedule; evaluate every arm from the original canonical-start state.

A matched Trust-Region Behavior Blending (TRB) arm is additionally required because it changes occupancy through a teacher-near behavior policy while retaining OPD; it is not yet implemented in the launcher. The method claim fails if random or progress-matched resets, TCOD-B2F, Guided-OPD, or TRB ties the targeted arm within the preregistered smallest effect of interest.

Player-state by history ablation. For the same certified player state, compare (i) a direct snapshot with the minimal canonical observation history, (ii) replay of a teacher trajectory with its authentic prefix, and (iii) a direct snapshot with a matched reconstructed history. A fourth Backplay condition anneals starts backward along a witness trajectory. This distinguishes initialization of latent player state x from information or error patterns carried in h, and tests whether a direct snapshot adds value beyond successful prefix replay. The present implementation does not restore a complete shared world (for example NPC, mob, resource, or concurrent-player state), so no full-world-state claim is made.

Weights by recovery factorial. Evaluate base, round-2, and round-3 weights with recovery both off and on. Preserve raw pre-rewrite emissions. These are conditional contrasts among three fixed checkpoint artifacts, not estimates of a training procedure. The registered full study uses 20 paired fresh-world replicate clusters, sums all three personality lanes before comparison, and freezes seven contrasts: round 2 and round 3 versus base with recovery off; recovery on versus off for each checkpoint; and both checkpoint-by-recovery interactions. It controls familywise alpha at .05 by a two-sided Bonferroni allocation. The 20-cluster record targets 80% power only under its prospective assumption of a three-stage minimum relevant paired difference and paired-difference standard deviation at most three; it is not an empirical variance estimate. The separate registered 18-cell local run is exploratory and cannot replace this design; no outcome enters the paper until its complete bundle passes the fail-closed audit.

Generalization. Hold out at least one quest from candidate discovery, state selection, and training, then compare full-walkthrough against no-walkthrough prompts. Report the probability of reaching the bottleneck, crossing it conditional on arrival, and finishing downstream. A second environment or model family is needed for any broad claim. Corrected-interface SFT and a SCoRe-style first-error-prefix condition are additional baselines under matched interaction and teacher- scoring budgets (Lyu et al., 2025).

Copy-prior test. Construct paired canonical/malformed histories across tools, states, defect types, and at least two teacher sizes or families. Score identical canonical and malformed continuations; report teacher and student log-odds with paired confidence intervals. Then intervene on grading context and test generation, rather than inferring generation from grader-side movement.

8 Related Work

GKD establishes on-policy distillation from student-generated sequences with teacher feedback (Agarwal et al., 2024); our work does not claim novelty for OPD itself. OPCD distills contextual experience on student-generated trajectories, including in text games (Ye et al., 2026); On-Policy Expert Corrections switch from student to expert within a trajectory to address covariate shift (Lauffer et al., 2025). TCOD directly studies temporal curricula and intermediate-state initialization for multi-turn OPD (Wang et al., 2026); broad novelty for seeded starts is therefore preempted. Guided-OPD changes live occupancy by mixing teacher and student turns (Li et al., 2026a); ReOPD designs reliable prefix distributions offline (Liao et al., 2026); TRB anneals a teacher-near rollout policy inside a student-centered trust region (Plyusov et al., 2026); and SCoRe localizes the first student error before short-horizon training (Lyu et al., 2025). SOD and KAT study supervision on degraded prefixes (Zhong et al., 2026; Xin et al., 2026). SOD in particular treats post-tool-error high-divergence steps as increasingly unreliable and reweights them; our unvalidated hypothesis is narrower, predicting locally wrong-signed candidate preference. Unmasking OPD likewise motivates per-token, per-context diagnostics rather than aggregate-loss interpretation (Armandpour et al., 2026). These methods make intermediate-state, student-failure, and state-centric novelty claims unavailable. DAgger formalizes iterative data collection under the learner's state distribution (Ross et al., 2011). A recent state-distribution analysis likewise argues that the source and locality of training states can matter as much as the loss (Nie, 2026). Reset-along-demonstration, Backplay, and Go-Explore show that restoring intermediate states can expose useful signal in hard-exploration reinforcement learning (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). We use the same control point for on-policy language-model distillation in a persistent, typed environment, but the current study does not establish superiority over those curriculum families.

The proposed state selector uses quest stage, inventory, position, and repeated teacher–student success gaps during training. This is privileged information and must be costed and evaluated for held-out transfer. Privileged Information Distillation and its on-policy self-distillation variant already study action-only transfer from privileged agent signals (Penaloza et al., 2026). SERL separately shows that the source and insertion point of environmental feedback matter in multi-turn agents (Li et al., 2026b); neither privilege nor feedback selection is novel here.

Game-agent benchmarks already cover long-horizon language and vision agents. BALROG emphasizes knowing–doing gaps across games (Paglieri et al., 2025), while Orak uses MCP across twelve games and releases gameplay fine-tuning data (Park et al., 2026). MCP and game play are therefore infrastructure here, not novelty claims. GITM uses LLM-generated plans, text memory, and structured actions in open-world Minecraft (Zhu et al., 2024). Voyager and AgentTrek use curricula, reusable skills, or guided replay to obtain long-horizon behavior (Wang et al., 2024; Xu et al., 2025). Our narrower question is whether a frozen selector over low-occupancy, teacher-competent persistent player states adds value beyond natural OPD, successful-prefix replay, turn-level guidance, and matched generic resets.

Supervised imitation can compound errors under distribution shift (Ross et al., 2011), and knowledge distillation does not guarantee functional agreement between teacher and student (Stanton et al., 2021). The r10 cross-vocabulary SFT regression in our development record is motivating evidence, but its raw artifact bundle is incomplete and it is not a primary result of this submission draft.

9 Conclusion

The audited case supports a concrete hypothesis: low-occupancy player states may hide useful teacher competence from natural OPD. Natural-visitation OPD changed behavior without changing score; a later hand-selected state-augmented round was followed by reported canonical-start passage of a wall that the earlier reported policies had not passed. That sequence does not establish the proposed selector. Random and progress-matched resets, TCOD-B2F, Guided-OPD, replication, and held-out transfer decide whether the narrow method survives. A malformed-prefix probe separately motivates, but does not yet establish, a wrong-signed teacher-feedback mechanism at the tool boundary. A prospective local pilot executed all registered cells with internally hash-checked evidence, but its short horizon produces no quest progress and no policy-quality claim.

Limitations

The central limitation is statistical: one run per principal arm cannot support a population claim, and three prompt variants are not independent seeds. The state-seeded round also differs from the preceding round in initialization and data. The environment is a single game with procedural walkthroughs, the teacher and student are from one model family, and stage counts are coarse. Some development analyses depend on logs whose raw inputs are not yet packaged. The paper must not be submitted as confirmatory until the study design above is run and a clean clone regenerates every main table.

Direct database construction is environment-specific and may not transfer to systems without restorable state. The historical failure state was selected retrospectively, so selection bias is possible; the prospective selector has no outcome yet. The local pilot has only three five-minute cells per snapshot; it was designed for feasibility, not power, and its outcomes cannot enter the confirmatory estimate. Teacher competence at the target state is observed in a small number of trajectories, not estimated precisely. The syntax probe covers one defect family and should not be generalized to all teacher-forced grading.

Ethical Considerations

The work concerns game agents acting in isolated local worlds. Risks include overstating autonomy, silently baking procedural knowledge into prompts, and mistaking recovered actions for model competence. We mitigate these by labeling scaffolding, logging model-visible schemas and raw emissions, separating runtime recovery from weights, and reporting failed interventions. No human-subject or private-user data are used.

References

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. 2024. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*.

Mohammadreza Armandpour, Fatih Ilhan, David Harrison, Ajay Jaiswal, Duc N. M. Hoang, Fartash Faghri, Yizhe Zhang, Minsik Cho, and Mehrdad Farajtabar. 2026. Unmasking on-policy distillation: Where it helps, where it hurts, and why. *arXiv preprint arXiv:2605.10889*.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. 2015. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*.

Adrien Ecoffet, Joost Huizinga, Joel Lehman, Kenneth O. Stanley, and Jeff Clune. 2021. First return, then explore. *Nature*, 590:580–586.

Kaetram contributors. 2026. Kaetram-open: An open-source 2d mmorpg. https://github.com/Kaetram/Kaetram-Open. Pinned project fork accessed 2026-07-23.

Niklas Lauffer, Xiang Deng, Srivatsa Kundurthy, Brad Kenstler, and Jeff Da. 2025. Imitation learning for multi-turn LM agents via on-policy expert corrections. *arXiv preprint arXiv:2512.14895*.

Gengsheng Li, Mao Zheng, Mingyang Song, Ruiqi Liu, Tianyu Yang, Jie Sun, Qiyong Zhong, Haiyun Guo, Junfeng Fang, Dan Zhang, and Jinqiao Wang. 2026a. On-policy distillation with curriculum turn-level guidance for multi-turn agents. *arXiv preprint arXiv:2606.15912*.

Xiaozhe Li, Tianyi Lyu, Yang Li, Yichuan Ma, Peiji Li, Linyang Li, Qipeng Guo, Dahua Lin, and Kai Chen. 2026b. What and when to distill: Selective hindsight distillation for multi-turn agents. *arXiv preprint arXiv:2605.19447*.

Baohao Liao, Hanze Dong, Christof Monz, Xinxing Xu, Li Dong, and Furu Wei. 2026. Multi-turn on-policy distillation with prefix replay. *arXiv preprint arXiv:2607.04763*.

Zizhuo Lin, Quanling Liu, Jinsheng Quan, Chao Zhang, Yifan Zhu, Xing Shi, Jingtao Xu, Zhihui Li, and Yawei Luo. 2026. Same evidence, different answers: Canonical-context on-policy distillation for multi-turn language models. *arXiv preprint arXiv:2605.30251*.

Yuanjie Lyu, Chengyu Wang, Jun Huang, and Tong Xu. 2025. From correction to mastery: Reinforced distillation of large language model agents. *arXiv preprint arXiv:2509.14257*.

Dong Nie. 2026. Post-training is about states, not tokens: A state distribution view of SFT, RL, and on-policy distillation. *arXiv preprint arXiv:2605.22731*.

Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, Jakob Foerster, Jack Parker-Holder, and Tim Rocktäschel. 2025. Balrog: Benchmarking agentic llm and vlm reasoning on games. In *International Conference on Learning Representations*.

Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu Park, Byeong-Uk Lee, Jaeyoung Hwang, Jaewoo Ahn, Ameya S. Mahabaleshwarkar, Bilal Kartal, Pritam Biswas, Yoshi Suhara, Kangwook Lee, and Jaewoong Cho. 2026. Orak: A foundational benchmark for training and evaluating llm agents on diverse video games. In *International Conference on Learning Representations*.

Emiliano Penaloza, Dheeraj Vattikonda, Nicolas Gontier, Alexandre Lacoste, Laurent Charlin, and Massimo Caccia. 2026. Privileged information distillation for language models. *arXiv preprint arXiv:2602.04942*.

Daniil Plyusov, Alexey Gorbatovski, Alexey Malakhov, Nikita Balagansky, Boris Shaposhnikov, Daria Korotyshova, and Daniil Gavrilov. 2026. Trust-region behavior blending for on-policy distillation. *arXiv preprint arXiv:2605.31159*.

Qwen Team. 2026. Qwen3.5-2B and Qwen3.5-4B model cards. https://huggingface.co/collections/Qwen/qwen35. Accessed 2026-07-23.

Cinjon Resnick, Roberta Raileanu, Sanyam Kapoor, Alexander Peysakhovich, Kyunghyun Cho, and Joan Bruna. 2018. Backplay: "man muss immer umkehren". *arXiv preprint arXiv:1807.06919*.

Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *International Conference on Artificial Intelligence and Statistics*.

Tim Salimans and Richard Chen. 2018. Learning montezuma's revenge from a single demonstration. *arXiv preprint arXiv:1812.03381*.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.

Samuel Stanton, Pavel Izmailov, Polina Kirichenko, Alexander A. Alemi, and Andrew Gordon Wilson. 2021. Does knowledge distillation really work? In *Advances in Neural Information Processing Systems*.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*.

Jiaqi Wang, Wenhao Zhang, Weijie Shi, Yaliang Li, and James Cheng. 2026. Tcod: Exploring temporal curriculum in on-policy distillation for multi-turn autonomous agents. *arXiv preprint arXiv:2604.24005*.

Haoran Xin, Anhao Zhao, Ying Sun, Jin Li, Xiaoyu Shen, and Hui Xiong. 2026. Escaping the kl agreement trap in on-policy distillation. *arXiv preprint arXiv:2606.09471*.

Yiheng Xu, Dunjie Lu, Zhennan Shen, Junli Wang, Zekun Wang, Yuchen Mao, Caiming Xiong, and Tao Yu. 2025. Agenttrek: Agent trajectory synthesis via guiding replay with web tutorials. In *International Conference on Learning Representations*.

Tianzhu Ye, Li Dong, Xun Wu, Shaohan Huang, and Furu Wei. 2026. On-policy context distillation for language models. *arXiv preprint arXiv:2602.12275*.

Qiyong Zhong, Mao Zheng, Mingyang Song, Xin Lin, Jie Sun, Houcheng Jiang, Xiang Wang, and Junfeng Fang. 2026. Sod: Step-wise on-policy distillation for small language model agents. *arXiv preprint arXiv:2605.07725*.

Xizhou Zhu, Yuntao Chen, Hao Tian, Chenxin Tao, Weijie Su, Chenyu Yang, Gao Huang, Bin Li, Lewei Lu, Xiaogang Wang, Yu Qiao, Zhaoxiang Zhang, and Jifeng Dai. 2024. Ghost in the minecraft: Hierarchical agents for minecraft via large language models with text-based knowledge and memory. In *International Conference on Learning Representations*.

A Training Details

The settings below are reconstructed from the current implementation and project notes committed after the June runs; they are reported settings, not a hash-attested historical training configuration.

The teacher is a scaffolded Qwen/Qwen3.5-4B model and the student begins from Qwen/Qwen3.5-2B (Qwen Team, 2026). Both use the same vocabulary. Student rollouts are reconstructed as the byte-exact serving prompt plus the verbatim emitted continuation. This prefix property avoids re-rendering differences caused by chat-template handling of reasoning delimiters. Teacher and behavior log probabilities are computed on the identical raw string. Every tenth session is held out for development gates.

The update uses PPO-style clipped importance weighting (Schulman et al., 2017) with clip $\epsilon = 0.3$, advantage clamp $[-3, 3]$, and a 1.5 weight on the first third of each session. Training uses one epoch, learning rate 5×10^{-5}, effective batch size 32, and a fresh LoRA initialized over the prior merged checkpoint. Action positions alone receive the language-model head. These choices were development decisions, not independently ablated.

Round 1 contains 5,564 train and 574 held-out records from the base policy's natural rollouts. Round 2 combines 5,523 natural round-1 records and 2,326 seeded records before the 7,024/825 train/held-out split. The seed encodes accepted Herbalist stage 1, Foraging level 5, starter inventory, and a walkability-checked tile near the target resource. Evaluation never loads this state.

B Claim-to-Evidence Boundary

Round-1 behavior. Descriptive, one run: measured behavior changed while score did not.

State-seeded round. Unreplicated: the seeded-training round was followed by canonical-start wall passage in one run.

Targeted selector. Protocol only: no candidate-state bundle, trained checkpoint, or outcome is currently available.

12 to 15. Reported recovery-off historical contrast; not an effect estimate or clean weights comparison, and exact parity remains unavailable.

18/30. Weights-plus-interface system result only.

Copy prior. Positive teacher-over-student advantage for one malformed continuation in a targeted historical probe; no matched preference reversal.

Recovery. Historical notes report no repeat marker after correction in rewritten sessions. Recovered rewritten bundles exist, but raw pre-rewrite emissions and an independently regenerated marker count are unavailable.

Local pilot. Nine short paired cells executed through the matched local stack with internally hash-checked artifacts; they provide no model-superiority or quest-performance claim and cannot enter the confirmatory estimate.

Historical bundles. The recovered external inventory binds the headline R10 and OPD agent-run directories and reproduces the R10 statistics, but it is not a public reviewer artifact and does not attest the executed reset command, game revision, or OPD training inputs.

Cross-task transfer. Unsupported; no claim.

C Reproducibility Contract

Before confirmatory runs, each immutable run bundle must contain: environment and game revisions; container or dependency locks; model, tokenizer, adapter, and dataset revisions; complete model-visible tool schemas; rendered-prompt hashes; sampling parameters; environment and inference seeds; raw emissions before any rewrite; state-transition logs; analysis outputs; and checksums. Dataset records must point to source runs and transformations. A clean-clone command must verify hashes and regenerate every table and figure without network-dependent mutable inputs.

The current development artifact does not yet meet this bar. A read-only external recovery now contains the headline R10 and OPD run directories and is bound to the SHA-256 digest of an external inventory. The checked-in run-digest audit independently hashes 60 selected directories totaling 18,812 files, and the R10 statistics regenerate from those paths. That recovery is not checked into this repository, publicly released, or sufficient to attest the historical reset command, full game revision, OPD training inputs, or exact render contract. From April 25 through PR #37's July 22 merge, the dedicated `run-eval.sh` lane on main reset a different database from its game servers (a fixed branch existed from July 18). The headline R10 and June OPD values used a separate 9001/9011/9021 orchestrator and are not implicated by that specific mismatch; the first model-visible state in all 36 recovered headline agent-runs matches the canonical level-1 start. Full dependency installation has now been demonstrated for the prospective CPU unit suite in a fresh clone, but not for training or paper-table regeneration. A separate local pilot has now executed nine matched hash-identified public-weight cells against a recorded game-fork commit, using one common Base tokenizer and patched render contract, registered inference/environment seeds, exact canonical-start projections, and 261 rehashed files in sealed per-cell inventories. This records prospective local game/model execution, not training, the full confirmatory factorial, historical results, or clean-clone table regeneration. Neither the recovered historical inventory nor the workstation-local pilot bundle is presently a reviewer-accessible artifact; release packaging, license review, and clean-clone regeneration remain submission gates. Confirmatory launch remains fail-closed. A draft isolated-service reachability checker executes hash-pinned MCP transitions and persistent player-state comparisons, but it has not yet produced a certificate in a live game/Mongo lane. A draft matched-training launcher freezes 50 core arm–seed cells plus 20 separately reported state–history ablation cells, shared budgets, interfaces, and held-out exclusion. A stacked draft adapter hash-verifies source material, checks held-out and arm-specific evidence, and emits normalized records marked `prepared_not_trained`; another stacked draft freezes a fresh bf16 LoRA ($r = 64$, $\alpha = 64$) over the same seven attention/MLP projections for every cell and provides a direct-token corrected-interface SFT execution path without re-rendering. A separate SCoRe-style adapter prepares correction SFT but fails closed before its short-horizon reward stage. A third draft validates prospective Guided-OPD records with complete-turn teacher/student roles, a frozen cosine training-step curriculum, and mixed-history traces; earlier stacked branches remain blocked from executing an unfaithful approximation. The live mixed-turn collector and actor-conditioned forward/reverse-KL backend remain unimplemented. Verified training bundles, SCoRe-style second-stage artifacts, and accelerator results also remain absent. These omissions are release blockers, not appendix footnotes.

D Planned Statistical Analysis

No session- or agent-level pseudo-replication will be used. For the causal training comparison, a freshly trained checkpoint is the independent unit and fresh-world evaluation seeds are nested repeated measurements. The primary method contrast must aggregate the frozen nested evaluations within each training seed before uncertainty across training seeds is estimated. The five seeds in the current launcher example are not a power justification.

For the fixed-checkpoint weights-by-recovery study, the independent unit is the paired fresh-world replicate cluster; inference is conditional on the three registered artifacts and cannot estimate training-seed variability. Its seven contrasts, 20 clusters, familywise error rule, and assumption-based power record are frozen prospectively. Before any method-outcome collection, the team must likewise preregister a smallest effect of interest and determine both levels of replication from independent pilot variance or a conservative prespecified variance grid. If variance is not defensibly known, use blinded variance re-estimation with a fixed maximum sample size. The primary comparison is two-sided unless a directional alternative and stopping rule are preregistered. Report all runs, exclusions, raw and standardized effects, uncertainty appropriate to the achieved design, and the complete secondary family with multiplicity handling. Exact wall-passage counts remain descriptive when sample sizes are small.